

# PROCEEDINGS OF THE NATIONAL CONFERENCE ON APPLICATIONS OF MATHEMATICS IN ENGINEERING SCIENCES

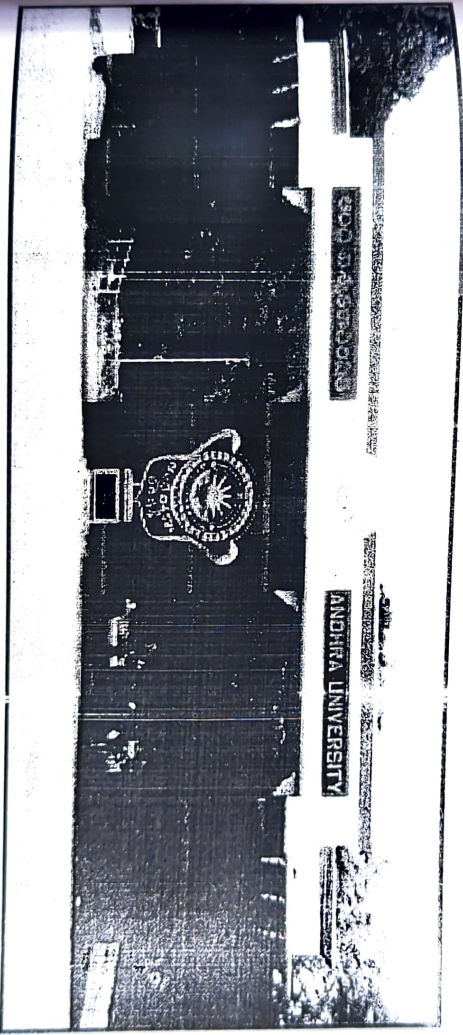
JUNE 29<sup>th</sup>-30<sup>th</sup>, 2011

ISBN 978-81-909851-0-7

## EDITORS

Prof. A.S.N. Murti

Dr. P. Vasudeva Reddy



DEPARTMENT OF ENGINEERING MATHEMATICS  
AU College Of Engineering (A)

Andhra University  
Visakhapatnam - 530 003  
Andhra Pradesh  
[www.andhrauniversity.info](http://www.andhrauniversity.info)

# Effects of Thermal Radiation and Magnetic Field on Unsteady Mixed Convection Flow and Mass Transfer Over a Porous Stretching Surface with I Generation

A. Balaji Ankit Reddy<sup>1, 2, 3</sup>, Sumeshan<sup>1</sup>, G.V. Rammanna Reddy<sup>1</sup>

<sup>1, 2, 3</sup> Department of Mathematics

<sup>1</sup> N.R.K.R.I.S.T. Tirtha Nagar, Nellore, A.P., India

<sup>2</sup> Usha Rama College of Engineering and Technology, A.P., India

<sup>3</sup> jayaramhsd@gmail.com, 9447120209@yahoo.co.in

<sup>2, 3</sup> S.R. Engineering College of Y.V.U, P.odduru, Kadapa, A.P., India

sumeshchayreddy@gmail.com

**Abstract:** The effects of thermal radiation and mass transfer on an unsteady hydromagnetic boundary layer mixed convection flow along a vertical porous stretching surface with heat generation is studied in this paper. The fluid is assumed to be viscous and incompressible. The governing partial differential equations are transformed into a system of ordinary differential equations using similarity variables. Numerical solutions of these equations are obtained by using Runge-Kutta fourth order method along with shooting technique. The behavior of the velocity, temperature, concentration, skin-friction coefficient, Nusselt number and Sherwood number for variations in the governing thermo physical parameters are computed, analyzed and discussed.

**Keywords:** Thermal radiation, MHD, stretching surface and heat generation.

## 1. INTRODUCTION

The study of boundary layer flow over a stretching surface has importance in the recent years due to its numerous industrial applications such as the aerodynamic extrusion of plastic sheets, the boundary layer along a liquid film, condensation process of metallic plate in a cooling bath and glass. Elbashbeshy and Bazid [1] presented an exact similarity solution for unsteady momentum and heat transfer flow whose motion is caused solely by the linear stretching of a horizontal stretching surface. Andersson *et al.* [2] presented a new similarity solution for the temperature field is devised, which transforms the time-dependent thermal energy equation to an ordinary differential equation. Nath *et al.* [3] obtained a set of similarity solutions for radiative-MHD stellar point explosion dynamics using shooting methods. In this chapter an attempt is made to investigate the effects of thermal radiation and magnetic field on unsteady mixed convection flow and mass transfer over a vertical stretching surface in the presence of heat source.

## II. MATHEMATICAL ANALYSIS

We consider an unsteady two-dimensional in convection boundary layer flow of an incompressible electrically conducting and radiating fluid over a vertical porous stretching surface moving with velocity

$U_w(x,t) = \frac{cx}{1-\alpha t}$ , where  $c$  and  $\alpha$  are constants and

temperature distribution  $T_w = T_\infty + \frac{T_0 e^{cx}}{2\alpha(1-\alpha t)}$

(Anjesson *et al.* [4]), where  $T_0$  is a reference temperature such that  $0 \leq T_0 \leq T_\infty$ . The  $x$ -axis is taken along the stretching surface in the direction of motion and the  $y$ -axis perpendicular to it. A uniform magnetic field of strength is applied normal to the stretching surface. The fluid is assumed to be gray, emitting and absorbing radiation, scattering medium. The Rosseland approximation is used to describe the radiative heat flux in the energy equation under the above assumptions, in the absence of an imposed electric field, the governing boundary layer equations

$$\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} = 0$$